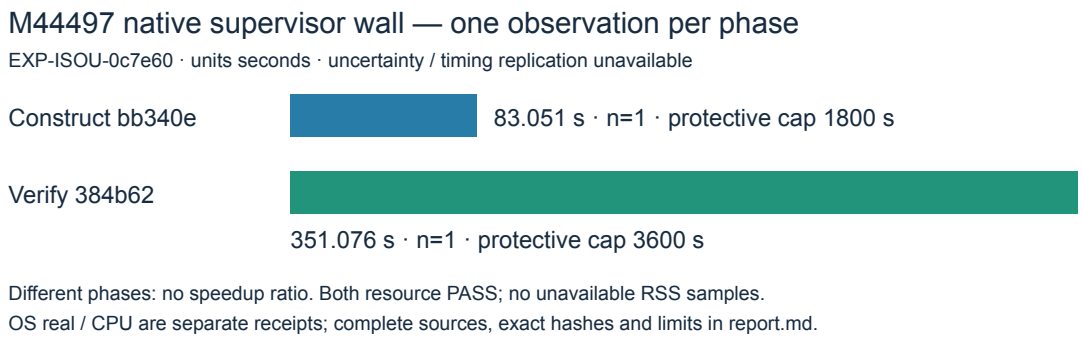
M44497: independent endpoint review

EV-ISOU-263837 / DEC-20261016-b60d77 / KN-FIND-14f145 Decision: support the selected compact known-CM self-isogeny endpoint on exact secp256k1. Degree 2^44497 − 1 (44,497 bits). The 54 expanded-map boards remain separate.



Certificate and independent review

Constructor CONSTRUCTED then fresh VERIFIED_COMPACT; certificate 63,615 bytes, SHA25685b5db36…b77f5d1. Thirteen entries hold: 11 PASS and 2 explicit trusted SEC2 qualifiers. Independent LL 44,495 transitions ends 0; Cornacchia 12,832 divisions; exact norm, endomorphism pairing, differential, eight fixed points and both dual compositions hold. Universal degree follows the reviewed endomorphism/dual theorem and unreduced norm a^2-ab+b^2. Point agreement alone is insufficient. IDC1hf56b3716b0a7eb8b has proof:null: finite supplement, separate from two exact-Z expansions and theorem review.

TASK 4bad89 report SHA 0a2f7014…11b371 holds both owned mathematical/resource joints. Review was raw-first and blind from producer/report conclusions; prior source contracts consulted only after rederivation. Requested xhigh, actual served model/effort unknown. Evidence strength replicated denotes independent arithmetic replay of this exact endpoint, not population/timing replication. Full 41,242-byte review and 22-path snapshot 5544d583…b349 are committed.

Cost and boundary

Native operations 83.029/351.050 s; supervisors 83.051/351.076 s. OS real 83.06/351.04 s; user 57.73/237.36 s; system 0.93/3.92 s; displayed derived CPU 58.66/241.28 s (total 299.94 s). Sampled worker peak 5,406,720/5,341,184 bytes, zero unavailable samples;8 GiB / 25 ms envelope with100 ms grace. Verifier OS real is 36 ms below supervisor: preserve separate instruments and unresolved difference. Shared M4 Pro, 14 logical CPUs, load 42–59, n=1 each: no A/A, uncertainty interval or speedup. Internal subphase costs unavailable; setup/build/control/custody costs excluded. Canonical CPU null unchanged.

Limits and next source work

SEC2 p/n primality is trusted, not a self-contained proof. No global record, novelty, security, prescribed-execution proof, external settlement or expanded-map admission. M8191 rejection/nonstart and P192 / 50341 censors remain.

Primary next: TASK de577e opt-in fixed-modulus reciprocal context across torsion and composition, reference/default unchanged; source/control/clean-build/custody and independent review before any benchmark/replay approval. Secondary bounds extension waits behind that gate. Goal remains active. Original run report and frontier figures checked unchanged. Full exact identities, resources/hashes, source theorems, metadata-failure appendix and references: report.md and original run report.